LangChain & RAG Beginner Guide

What is LangChain?

LangChain is a framework for building LLM applications. It connects LLMs with prompts, documents, tools, memory, and retrieval.

Core Components

Models, Prompts, Output Parssers, Document Loaders, Text Splitters, Embeddings, Vector Stores, Retrievers, Chains.

RAG

Retrieval-Augmented Generation retrieves relevant documents from a knowledge base before asking the LLM to answer.

RAG Pipeline

Load documents -> Split into chunks -> Create embeddings -> Store in vector database -> Retrieve relevant chunks -> Send context + question to LLM -> Generate answer.

Document Loaders

PyPDFLoader, TextLoader, CSVLoader, WebBaseLoader.

Text Splitters

RecursiveCharacterTextSplitter is commonly used to create overlapping chunks.

Embeddings

Convert text into vectors so semantic similarity can be searched.

Vector Stores

Examples: FAISS, Chroma, Pinecone, Weaviate, Milvus.

Retriever

Fetches the most relevant chunks using similarity search.

Simple LangChain Flow

User -> Prompt -> LLM -> Response. In RAG, retrieved context is inserted into the prompt.

Interview Questions

1. What is RAG? 2. Difference between embeddings and vector databases? 3. Why chunk documents? 4. What is similarity search?